MEMORY CONTROLLER SPECIFICATION

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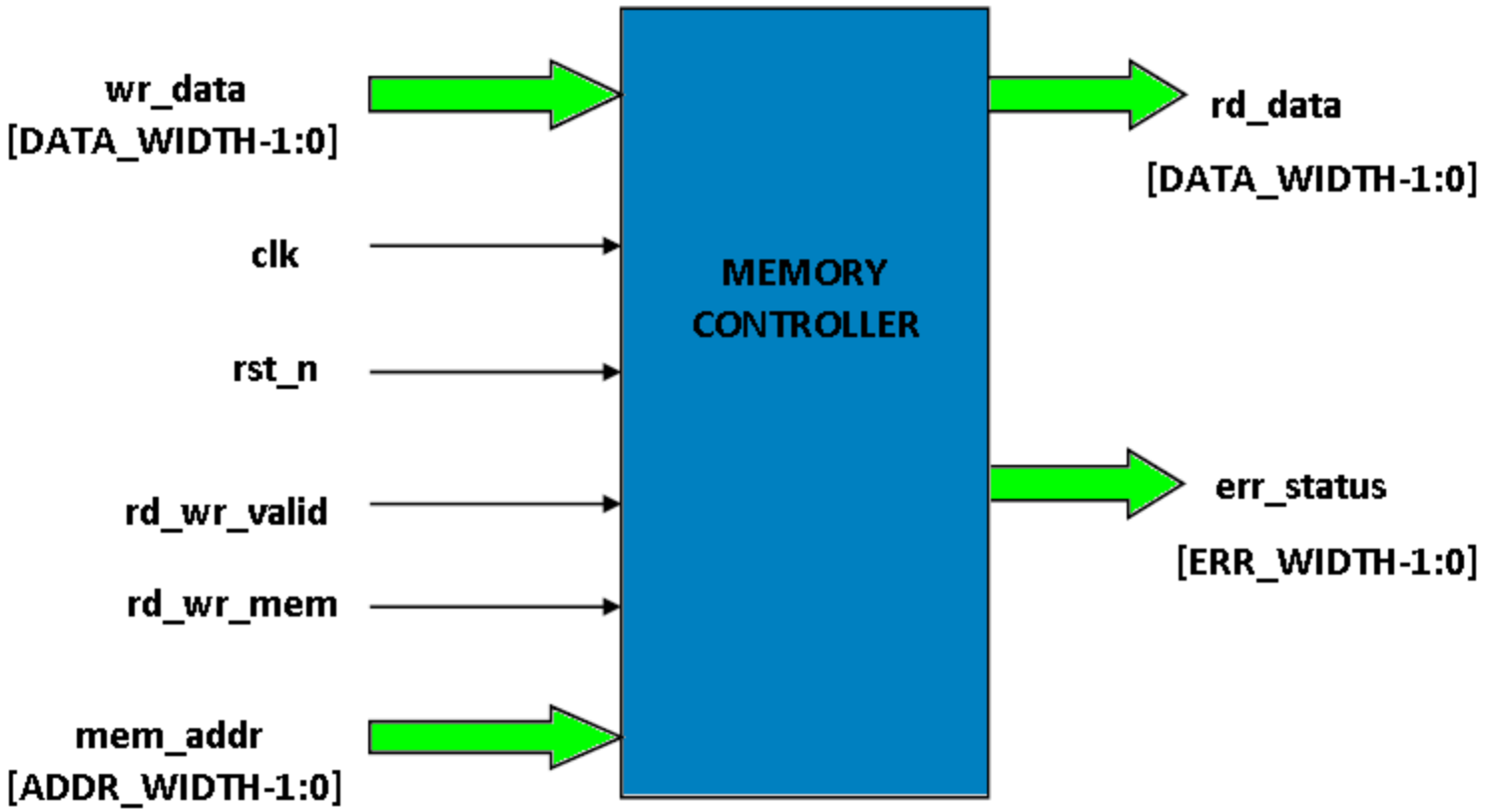
BLOCK DIAGRAM:



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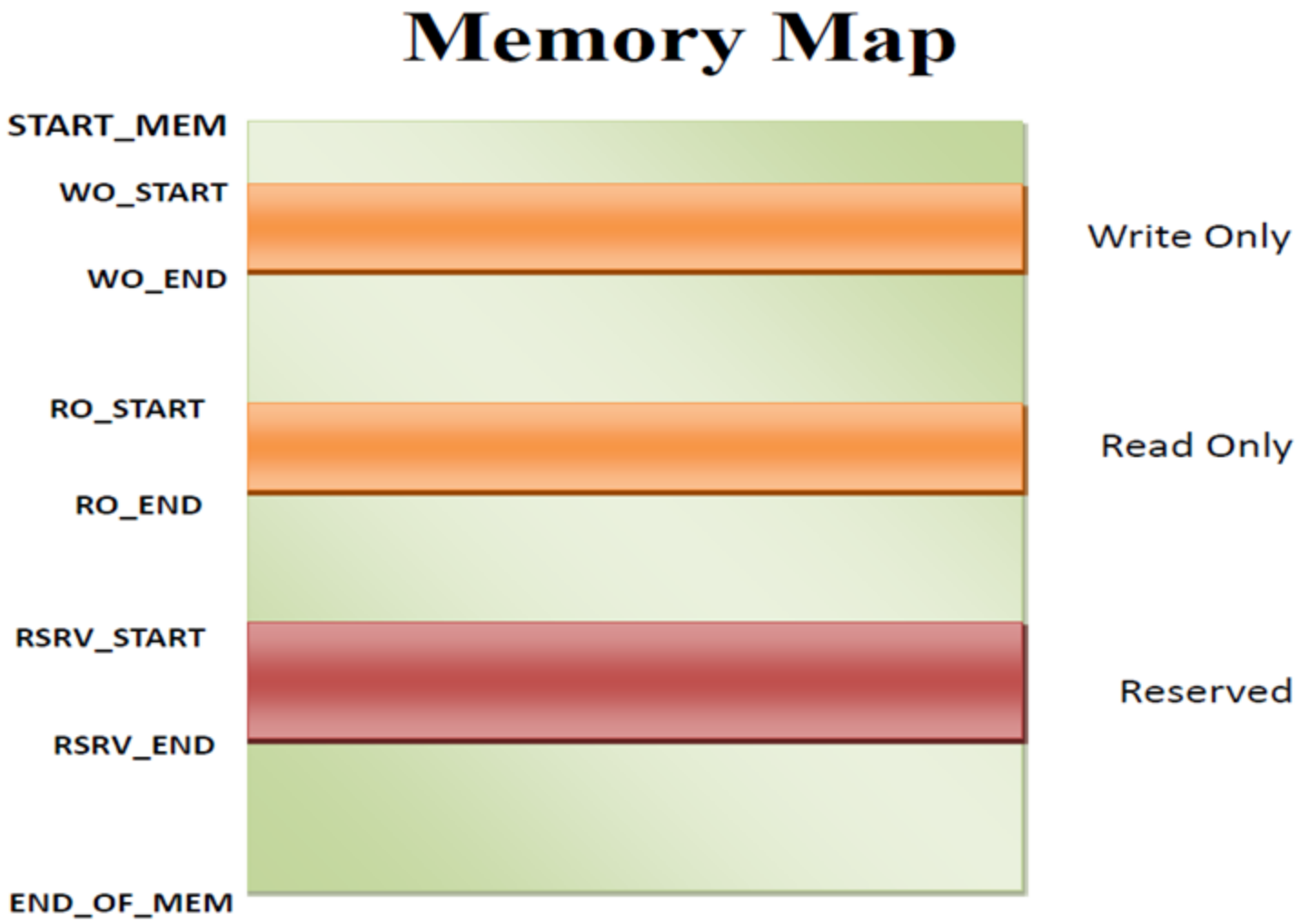
MEMORY MAP:



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MEMORY MAP:

DEPTH OF MEMORY = 2048

WO_START = 64
WO_END = 319

RO_START = 960
RO_END = 1215

RSRV_START = 1280
RSRV_END = 1343

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SIGNAL DESCRIPTION:

SIGNAL	DIRECTION	WIDTH	DESCRIPTION
clk	Input	1 bit	Clock input
rst_n	Input	1 bit	Active low reset input
wr_data [DATA_WIDTH-1:0]	Input	8 bits	Write data
rd_wr_mem	Input	1 bit	Indicates read/write operation
rd_wr_valid	Input	1 bit	Indicates read/write valid
mem_addr [ADDR_WIDTH-1:0]	Input	11 bits	Indicates read/write memory address
rd_data [DATA_WIDTH-1:0]	Output	8 bits	Read data
err_status [ERR_WIDTH-1:0]	Output	3 bits	Error signal

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DESCRIPTION:

a) When “rst_n” is low, “rd_data” is zero and “err_status” is also zero.

b) When both “rst_n” & “rd_wr_valid” are high then write operation takes place if “rd_wr_mem” is high and read operation takes place if “rd_wr_mem” is low.

c) When “rd_wr_valid” is low, no read/write operation takes place.

d) There is no clock latency for read/write operations.

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DESCRIPTION:

e) "err_status" signal indicates invalid access as given below:

err_status = 3'b000 => no error

err_status = 3'b001 => read in write only error

err_status = 3'b010 => write in read only error

err_status = 3'b100 => reserved access error

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PARAMETERS:

DATA_WIDTH = 8

ADDR_WIDTH = 11

ERR_WIDTH = 3

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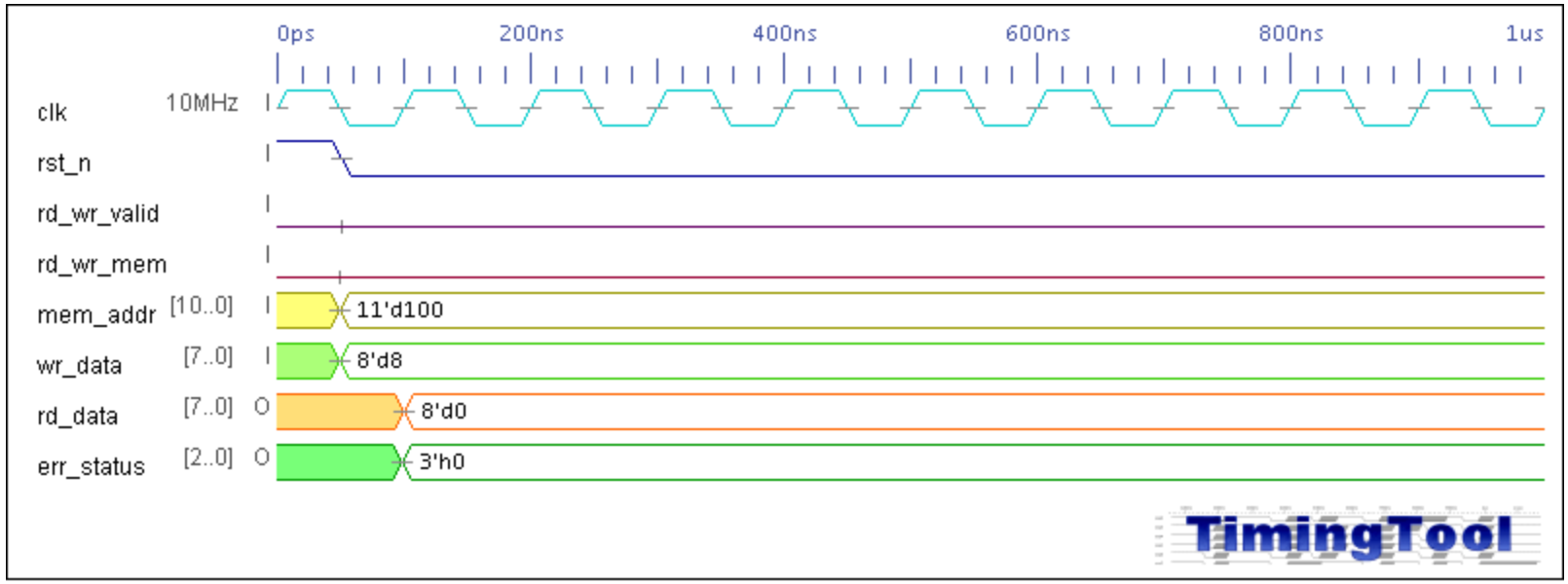
TIMING DIAGRAMS:

1) Reset operation



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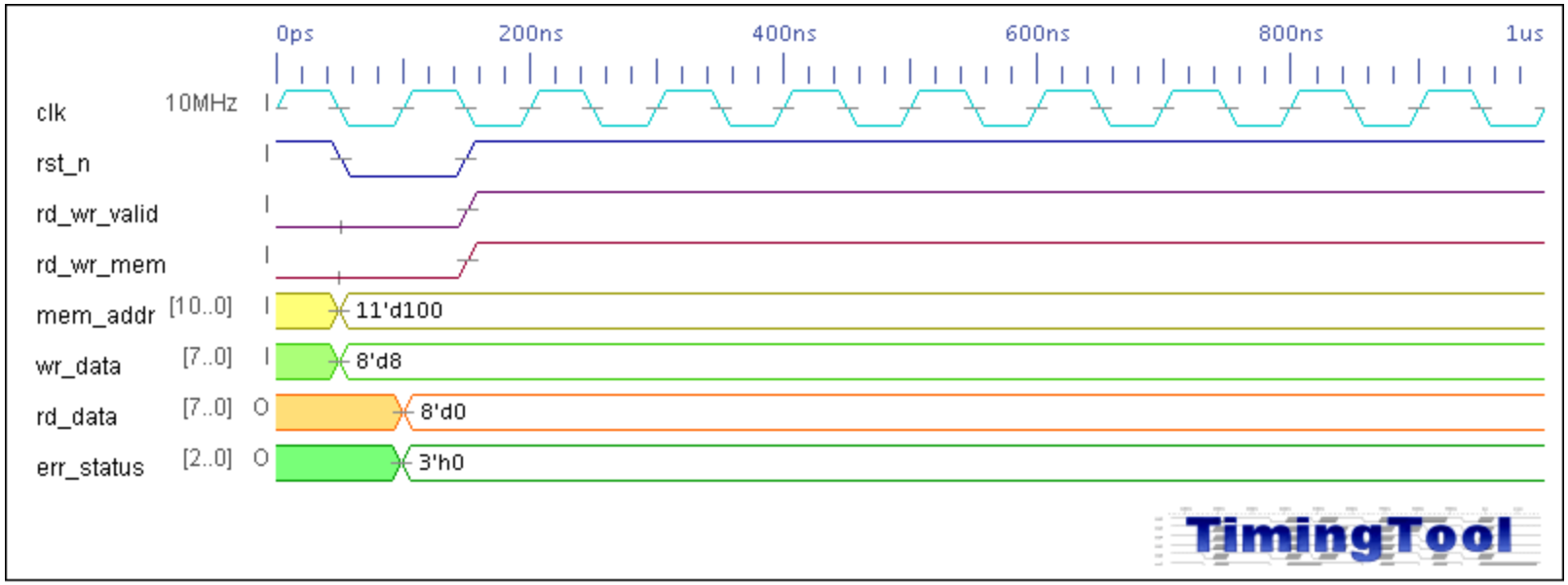
TIMING DIAGRAMS:

2) Write operation



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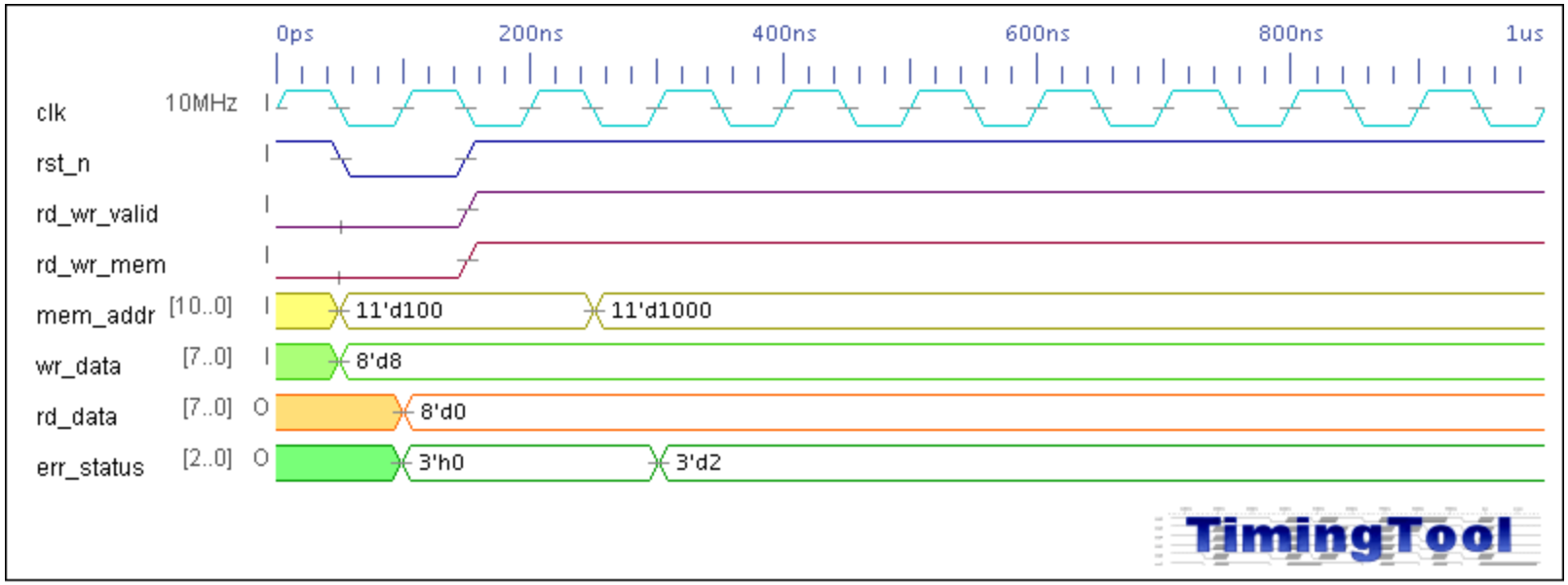
TIMING DIAGRAMS:

3) Write operation in RO space



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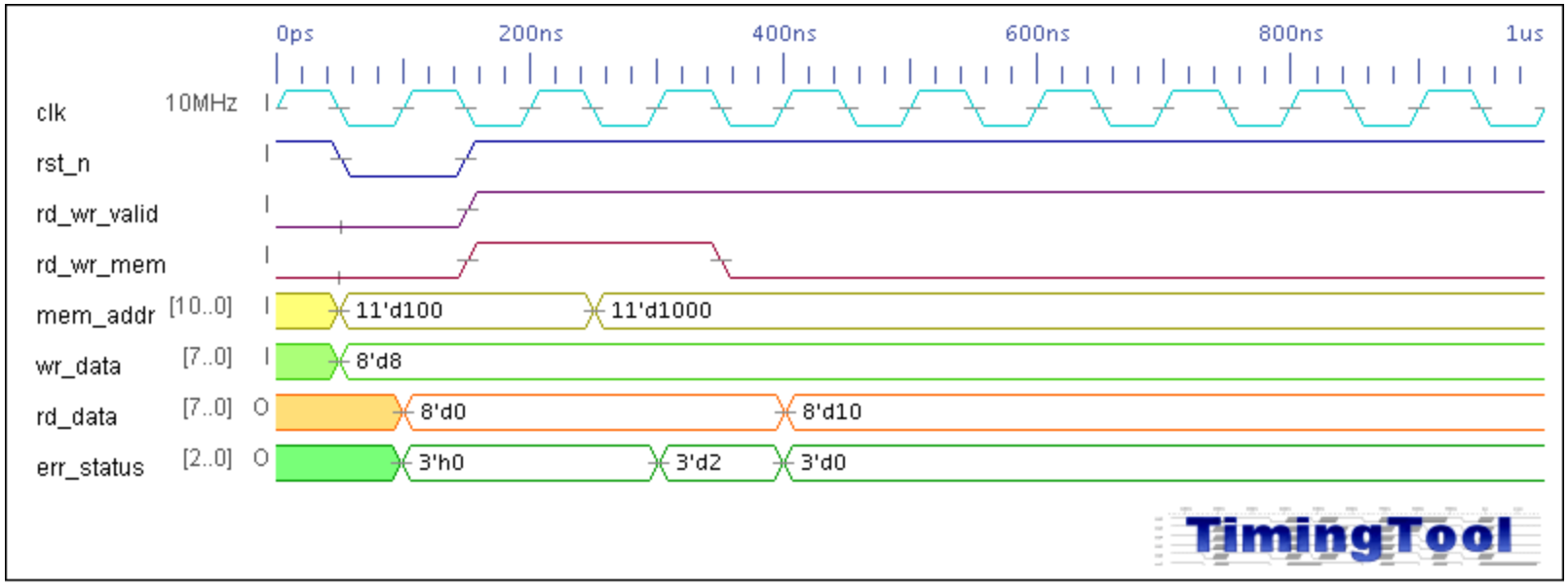
TIMING DIAGRAMS:

4) Read operation



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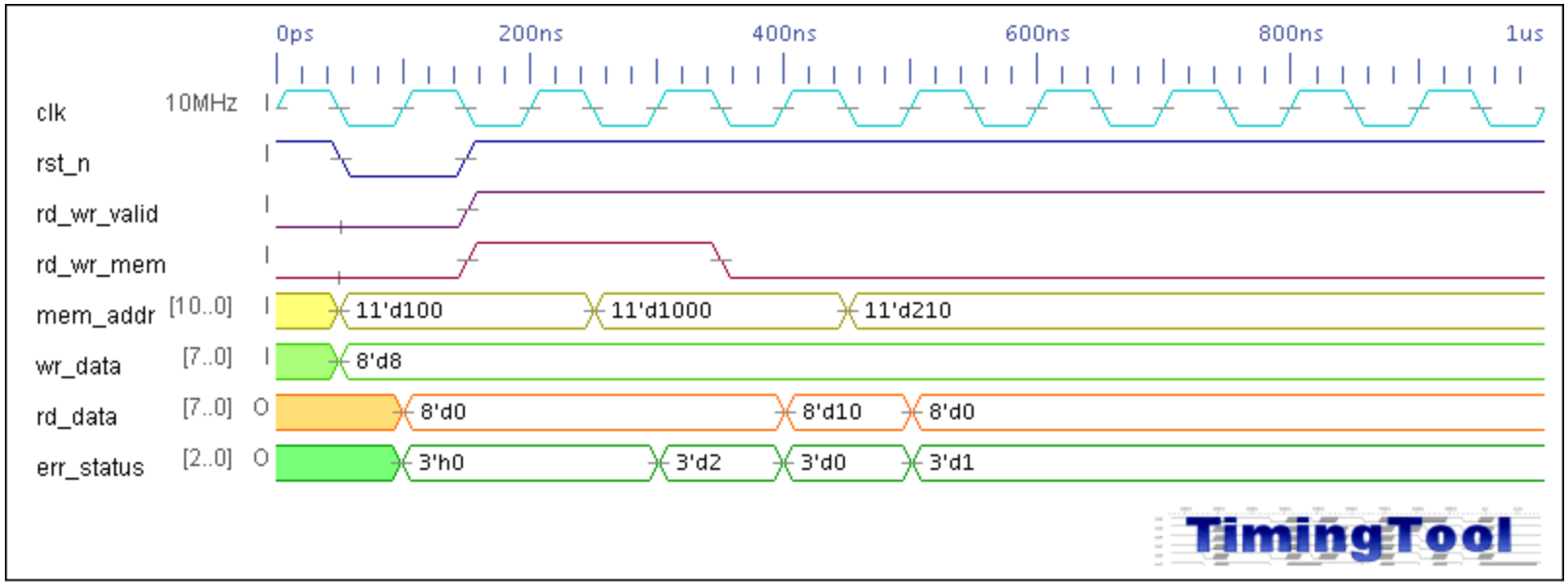
TIMING DIAGRAMS:

5) Read operation in WO space



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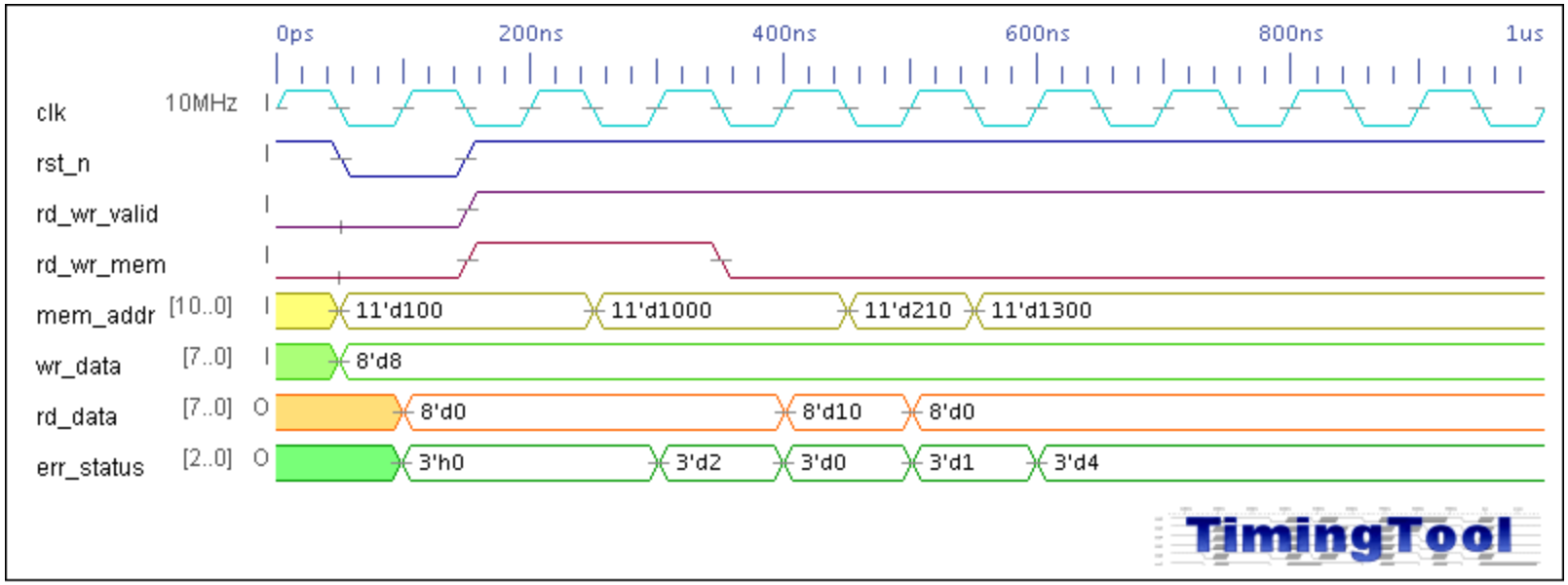
TIMING DIAGRAMS:

6) Read operation in reserved space



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